cent have no charger at all, and 14 per cent share one. This disparity is a huge pain point, particularly in cities, where giant apartment complexes mean you can't instal a charger of your own.

Even worse is the regional discrepancy in charging infrastructure, says Abhinav Kalia, CEO of ARC Electric, a B2B EV cab platform. EV infrastructure is nearly completely absent in rural, remote and hilly areas, where cleaner forms of mobility could potentially have the biggest impact. Be it the delivery EV in a Tier 3 town or public transport EVs in a border district, the lack of fast and reliable charging infrastructures is preventing scaling these solutions. These are not just geographical blind spots but barriers to equitable EV adoption, says Kalia.

Running Out of Charge

Range anxiety—will my EV run for the entire length of my journey, and will I get a charging point nearby if it runs low on charge—adds a layer of apprehension. "I can go to Chandigarh from Delhi without any issues. But, for the hills, I'm not so sure," says Ramanjit, an MNC executive who owns a mini-SUV with a proclaimed range of 350 km. "The highways generally do not have reliable charging options yet. Even hotels are not conducive to EV owners' needs."

Charging stations are also not immune to the way Indians generally treat public infrastructure. In an August 2024 study by the Institute of Energy Economics and Financial Analysis study, 31 out of 37 chargers (83.7 percent) in Delhi were found non-functional—dead display screens, missing connectors, no power supply. They also found that roughly three-quarters of inspected chargers used the first-generation Bharat DC-001 connector, now largely incompatible with modern CCS2-enabled cars; that low utilisation feeds a vicious circle of neglect and vandalism.

The study found charging speed to be an issue as well. While 'fast charging' remained the desirable option for the majority, Indian consumers came out as more patient than in other geographies, with 69 per cent prepared to spend 40 minutes or more for an 80 per cent charge at public stations. Forty one per cent say 21 to 40 minutes is reasonable, 23 per cent are okay with 11 to 20 minutes, and a demanding 5 per cent will tolerate only 10 minutes or less. Under the PM E-DRIVE, thankfully, 22,100 fast chargers are slated to come up.

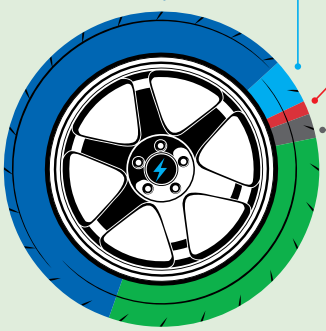
In comparison, megawatt chargers from BYD, Xpeng and Zeekr can add 200 to 400 km of range in under 10 minutes in China, even as battery-swap stations for two- and three-wheelers are spreading rapidly in dense urban markets. Europe and the US are pushing for integrated networks that blend fast, destination and home charging to improve reliability and geographic equity. "Technology has to provide the answers in the long run," says Shenu Agrawal, vice-president of the Society of Indian Automobile Manufacturers. "We have to have a minimum of 300 km range; charging time and cost have to decrease. Until product solutions are available, we need to find an ecosystem solution." As of now, these measures remain a pipe dream for India, even if the destination is in sight. But a crawl won't do—this journey demands a full throttle. ■

PENETRATION OF EVs IN INDIA

While two-wheeler EVs remain the most popular choice in India, three- and four-wheelers have also found takers, denoted by a 10.5 per cent and 18 per cent YoY increase in registrations in FY24-25, respectively

Electric two-wheelers (1.1% of all two-wheelers)
(scooters and motorcycles)
3.2–3.5 MILLION
56–62% of all EVs

Electric three-wheelers (15% of all three-wheelers)
(e-rickshaws, which are a majority, and auto-rickshaws)
1.8–2 MILLION
32–35% of all EVs



Electric light four-wheelers (0.3% of all light four-wheelers)
(cars and SUVs)
250,000–300,000
4–5% of all EVs

Electric medium/ heavy four-wheelers (0.9% of all medium/ heavy four-wheelers)
(buses and trucks)
50,000–100,000
1–2% of all EVs

Other EVs
(including low-speed vehicles, e-carts, specialised vehicles)
120,000
2–3% of all EVs

Sources: SIAM; Ministry of Road Transport and Highways



"Technology has to give answers in the long run. We need a 300 km range, minimum. Charging time has to decrease"

SHENU AGRAWAL
Vice-President, Society of Indian Automobile Manufacturers (SIAM)